



Πανεπιστήμιο Πειραιώς
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Τμήμα Ψηφιακών Συστημάτων

Επίπεδο: Προπτυχιακό Πρόγραμμα Σπουδών

Ασφάλεια Δικτύων

Lab exercise 3

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Instructions

The goal of this lab is to configure the DHCP Server and R1, R2 and R3 to allow for dynamic IP leasing for the networks 10.0.0.0 /24, 20.0.0.0 /24, 30.0.0.0 /24, 40.0.0.0 /24 and 50.0.0.0 /24 respectively:

- o Change the hostnames of R1, R2, R4, R5 and R6.
- o Configure Routers interfaces (g0/0, g0/1, g0/2) with the proper IPs and enable OSPF.
- o Configure the *ip helper-address* interface command on the interfaces of the routers so that everyone gets an IP from the DHCP Server.
- o Configure the DHCP pools on the DHCP Server for the networks 10.0.0.0 /24, 20.0.0.0 /24, 30.0.0.0 /24, 40.0.0.0 /24 and 50.0.0.0 /24
- o Configure the 10.0.0.0 /24 pool so that the Admin gets the IP 10.0.0.100.
- o Configure the 20.0.0.0 /24 pool so that Alice gets the IP 20.0.0.200.
- o Configure the 30.0.0.0/24 pool so that only the 5 Workers PCs can take ip addresses.

- o Configure the 50.0.0.0/24 pool so that CEO can take ip address 50.0.0.25 .
- o Configure the 40.0.0.0/24 pool so that Receptionist can take ip address 40.0.0.65 .

Once DHCP is configured, on the Desktop Tab of each PC in the IP Configuration window choose DHCP. Every PC should be leased an IP proper to its network location.

Step 1: Changing routers' hostnames:

In order to accomplish all our steps, we must select each router, select the CLI, enter as admin using the command “en”, type “conf t”. To complete our first step we must type the command: “hostname <<name>>” where name is the hostname of the router we want to change.

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router_Reception
Router_Reception(config)#|
```

Figure 1: Changing the hostname of reception's router

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router_AdminRoom
Router_AdminRoom(config)#|
```

Figure 2: Changing the hostname of admin room's router

```
ChangeMe>en
ChangeMe#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ChangeMe(config)#hostname Router_BranchOffice
Router_BranchOffice(config)#|
```

Figure 3: Changing the hostname of branch office's router

```
ChangeMe>en
ChangeMe#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ChangeMe(config)#hostname Router_WorkerRoom
Router_WorkerRoom(config)#|
```

Figure 4: Changing the hostname of worker room's router

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router_EcxecutiveRoom
Router_EcxecutiveRoom(config)#
```

Figure 5: Changing the hostname of executive room's router

```
R4>en
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#hostname Router_link
Router_link(config)#
```

Figure 6: Changing the hostname of R4 router

Step 2: Configure Routers Interfaces

To configure our router, we must select the interface we want to configure (g0/0, g0/1, g0/2) using the command `int`. Then, we use the “ip address” command specifying the ip and the subnet mask. Next, to save the running configuration, we must type “do write meme” and lastly, to activate the network interfaces we must type “no shut”.

```
Router_Reception>en
Router_Reception#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_Reception(config)#int g0/0
Router_Reception(config-if)#ip address 40.0.0.1 255.255.255.0
Router_Reception(config-if)#do write mem
Building configuration...
[OK]
Router_Reception(config-if)#no shut
Router_Reception(config-if)#exit
Router_Reception(config)#int g0/1
Router_Reception(config-if)#ip address 172.3.4.2 255.255.255.252
Router_Reception(config-if)#do write mem
Building configuration...
[OK]
Router_Reception(config-if)#no shut
Router_Reception(config-if)#exit
Router_Reception(config)#int g0/2
Router_Reception(config-if)#ip address 192.168.1.1 255.255.255.0
Router_Reception(config-if)#do write mem
Building configuration...
[OK]
Router_Reception(config-if)#no shut

Router_Reception(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

Figure 7: Configure all interfaces of Reception's Router

```

Router_AdminRoom>en
Router_AdminRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_AdminRoom(config)#int g0/0
Router_AdminRoom(config-if)#ip address 10.0.0.1 255.255.255.0
Router_AdminRoom(config-if)#do write mem
Building configuration...
[OK]
Router_AdminRoom(config-if)#no shut

Router_AdminRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router_AdminRoom(config)#int g0/1
Router_AdminRoom(config-if)#ip address 172.1.4.1 255.255.255.252
Router_AdminRoom(config-if)#do write mem
Building configuration...
[OK]
Router_AdminRoom(config-if)#no shut

Router_AdminRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router_AdminRoom(config)#int g0/2
Router_AdminRoom(config-if)#ip address 172.3.4.1 255.255.255.252
Router_AdminRoom(config-if)#do write mem
Building configuration...
[OK]
Router_AdminRoom(config-if)#no shut

Router_AdminRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
|

```

Figure 8: Configure all interfaces of AdminRoom's router

```

Router_link>en
Router_link#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_link(config)#int g0/0
Router_link(config-if)#ip address 172.1.5.1 255.255.255.252
Router_link(config-if)#do write mem
Building configuration...
[OK]
Router_link(config-if)#no shut

Router_link(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router_link(config)#int g0/1
Router_link(config-if)#ip address 172.1.4.2 255.255.255.252
Router_link(config-if)#do write mem
Building configuration...
[OK]
Router_link(config-if)#no shut
Router_link(config-if)#exit
Router_link(config)#int
01:11:14: %OSPF-5-ADJCHG: Process 100, Nbr 10.0.0.1 on GigabitEthernet0/1 :
Loading Done
g0/2
Router_link(config-if)#ip address 172.4.3.2 255.255.255.252
Router_link(config-if)#do write mem
Building configuration...
[OK]
Router_link(config-if)#no shut

Router_link(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

```

Figure 9: Configure all interfaces of router we use to link other routers

```

Router_BranchOffice>en
Router_BranchOffice#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_BranchOffice(config)#int g0/0
Router_BranchOffice(config-if)#ip address 172.3.2.2 255.255.255.252
Router_BranchOffice(config-if)#do write mem
Building configuration...
[OK]
Router_BranchOffice(config-if)#no shut

Router_BranchOffice(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router_BranchOffice(config)#int g0/1
Router_BranchOffice(config-if)# 20.0.0.1 255.255.255.0
^
% Invalid input detected at '^' marker.

Router_BranchOffice(config-if)#ip address 20.0.0.1 255.255.255.0
Router_BranchOffice(config-if)#do write mem
Building configuration...
[OK]
Router_BranchOffice(config-if)#no shut

Router_BranchOffice(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router_BranchOffice(config-if)#exit
Router_BranchOffice(config)#int g0/2
Router_BranchOffice(config-if)#ip address 172.4.3.1 255.255.255.252
Router_BranchOffice(config-if)#do write mem
Building configuration...
[OK]
Router_BranchOffice(config-if)#no shut

Router_BranchOffice(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

```

Figure 10: Configure all interfaces of BranchOffice's router

```

Router_WorkerRoom>en
Router_WorkerRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_WorkerRoom(config)#int g0/0
Router_WorkerRoom(config-if)#ip address 30.0.0.1 255.255.255.0
Router_WorkerRoom(config-if)#do write mem
Building configuration...
[OK]
Router_WorkerRoom(config-if)#no shut

Router_WorkerRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router_WorkerRoom(config)#int g0/1
Router_WorkerRoom(config-if)#ip address 172.2.5.1 255.255.255.252
Router_WorkerRoom(config-if)#do write mem
Building configuration...
[OK]
Router_WorkerRoom(config-if)#no shut

Router_WorkerRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router_WorkerRoom(config)#int g0/2
Router_WorkerRoom(config-if)#ip address 172.3.2.1 255.255.255.252
Router_WorkerRoom(config-if)#do write mem
Building configuration...
[OK]
Router_WorkerRoom(config-if)#no shut

Router_WorkerRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

```

Figure 11Configure all interfaces of WorkerRoom's router


```

Router_EcutiveRoom>en
Router_EcutiveRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_EcutiveRoom(config)#int g0/0
Router_EcutiveRoom(config-if)#ip address 172.1.5.2 255.255.255.252
Router_EcutiveRoom(config-if)#do write mem
Building configuration...
[OK]
Router_EcutiveRoom(config-if)#no shut

Router_EcutiveRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router_EcutiveRoom(config)#int g0/1
Router_EcutiveRoom(config-if)#172.2.5.2 255.255.255.252
      ^
% Invalid input detected at '^' marker.

Router_EcutiveRoom(config-if)#ip address 172.2.5.2 255.255.255.252
Router_EcutiveRoom(config-if)#do write mem
Building configuration...
[OK]
Router_EcutiveRoom(config-if)#no shut

Router_EcutiveRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router_EcutiveRoom(config)#int g0/2
Router_EcutiveRoom(config-if)#ip address 50.0.0.1 255.255.255.0
Router_EcutiveRoom(config-if)#do write mem
Building configuration...
[OK]
Router_EcutiveRoom(config-if)#no shut

Router_EcutiveRoom(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
|

```

Figure 12: Configure all interfaces of ExecutiveRoom's router

Step 3: Enable OSPF

OSPF (Open Shortest Path First) is a link-state routing protocol used in IP networks to dynamically find the shortest and most efficient path for data between routers. It operates within a hierarchical structure, using areas to efficiently share routing information and adapt to network changes quickly. To enable it, we start the process by typing `OSPF 100`. Then we use the command “`network`” specifying the network address, wildcard mask, and area (typically area 0), where wildcard mask is the opposite of the subnet mask (eg. When subnet mask is 255.255.255.0, the wildcard mask is 0.0.0.255).

```

Router_Reception>en
Router_Reception#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_Reception(config)#router ospf 100
Router_Reception(config-router)#network 40.0.0.0 0.0.0.255 area 0
Router_Reception(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router_Reception(config-router)#network 172.3.4.0 0.0.0.3 area 0
Router_Reception(config-router)#exit

```

Figure 13: Enable OSPF in reception's router

```

Router_AdminRoom>en
Router_AdminRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_AdminRoom(config)#router ospf 100
Router_AdminRoom(config-router)#network 172.3.4.0 0.0.0.3 area 0
Router_AdminRoom(config-router)#network
02:14:08: %OSPF-5-ADJCHG: Process 100, Nbr 192.168.1.1 on GigabitEthernet0/2 from LOADING to FULL,
Loadi
Router_AdminRoom(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router_AdminRoom(config-router)#network 172.1.4.0 0.0.0.3 area 0
Router_AdminRoom(config-router)#exit

```

Figure 14: Enable OSPF in AdminRoom's router

```

Router_link>en
Router_link#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_link(config)#router ospf 100
Router_link(config-router)#network 172.1.4.0 0.0.0.3 area 0
Router_link(config-router)#network 172.1.5.0 0.0.0.3 area 0
Router_link(config-router)#network 172.4.3.0 0.0.0.3 area 0
Router_link(config-router)#exit
Router_link(config)#

```

Figure 15: Enable OSPF in router we use to link other routers

```

Router_BranchOffice>en
Router_BranchOffice#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_BranchOffice(config)#router ospf 100
Router_BranchOffice(config-router)#network 172.4.3.0 0.0.0.3 area 0
Router_BranchOffice(config-router)#network 172.3.2.0 0.0.0.3 area 0
Router_BranchOffice(config-router)#network 20.0.0.0 0.0.0.255 area 0
Router_BranchOffice(config-router)#exit

```

Figure 16: Enable OSPF in BranchOffice's router

```

Router_WorkerRoom>en
Router_WorkerRoom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router_WorkerRoom(config)#router ospf 100
Router_WorkerRoom(config-router)#network 172.3.2.0 0.0.0.3 area 0
Router_WorkerRoom(config-router)#network 172.2.5.0 0.0.0.3 area 0
Router_WorkerRoom(config-router)#network 30.0.0.0 0.0.0.255 area 0
Router_WorkerRoom(config-router)#exit

```

Figure 17: Enable OSPF in WorkerRoom's router

```

Router_EcutiveRoom>en
Router_EcutiveRoom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router_EcutiveRoom(config)#router ospf 100
Router_EcutiveRoom(config-router)#network 50.0.0.0 0.0.0.255 area 0
Router_EcutiveRoom(config-router)#network 172.1.5.0 0.0.0.3 area 0
Router_EcutiveRoom(config-router)#network 172.2.5.0 0.0.0.3 area 0
Router_EcutiveRoom(config-router)#exit

```

Figure 18: Enable OSPF in ExecutiveRoom's router

Step 4: Configure the ip helper-address interface

The ip helper-address command in Cisco networking is used to enable a router to forward **DHCP (Dynamic Host Configuration Protocol) requests** from clients on one subnet to a DHCP server on a different subnet. When configured on a router interface, this command tells the router to listen for DHCP broadcast requests on that interface and then forward them as unicast packets to the specified DHCP server's IP address.

```

Router_AdminRoom>en
Router_AdminRoom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router_AdminRoom(config)#int g0/0
Router_AdminRoom(config-if)#ip helper-address 192.168.1.100
Router_AdminRoom(config-if)#do write mem
Building configuration...
[OK]
Router_AdminRoom(config-if)#

```

Figure 19: ip helper-address so Admining gets the IP 10.0.0.100

```

Router_BranchOffice>en
Router_BranchOffice#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_BranchOffice(config)#int g0/1
Router_BranchOffice(config-if)#ip helper-address 192.168.1.100
Router_BranchOffice(config-if)#do write mem
Building configuration...
[OK]
Router_BranchOffice(config-if)#

```

Figure 20: ip helper-address so Alice gets the IP 20.0.0.200

```

Router_WorkerRoom>en
Router_WorkerRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_WorkerRoom(config)#int g0/0
Router_WorkerRoom(config-if)#ip helper-address 192.168.1.100
Router_WorkerRoom(config-if)#do write mem
Building configuration...
[OK]
Router_WorkerRoom(config-if)#

```

Figure 21: Configure the ip helper-address so only 5 workers can take IP's

```

Router_Reception>en
Router_Reception#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_Reception(config)#int g0/0
Router_Reception(config-if)#ip helper-address 192.168.1.100
Router_Reception(config-if)#do write mem
Building configuration...
[OK]
Router_Reception(config-if)#

```

Figure 22: Configure the ip helper-address so the receptionist takes the ip address 40.0.0.65

```

Router_EcxecutiveRoom>en
Router_EcxecutiveRoom#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router_EcxecutiveRoom(config)#int g0/2
Router_EcxecutiveRoom(config-if)#ip helper-address 192.168.1.100
Router_EcxecutiveRoom(config-if)#do write mem
Building configuration...
[OK]
Router_EcxecutiveRoom(config-if)#

```

Figure 23: Configure the ip helper-address so CEO can take ip address 50.0.0.25

Step 5: Configure the DHCP pools

To configure the DHCP pools we must select the server, go to services and select the DHCP. Then we turn on the service and we fill the pool name, default gateway, start ip address, subnet mask, and the maximum number of users and click save.

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: executivePool

Default Gateway: 50.0.0.1

DNS Server: 0.0.0.0

Start IP Address: 50.0.0.25

Subnet Mask: 255.255.255.0

Maximum Number of Users: 10

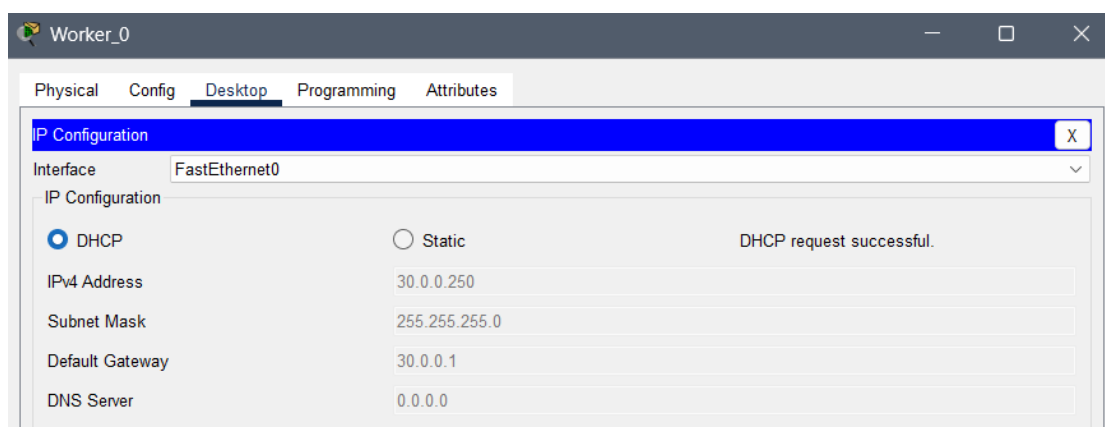
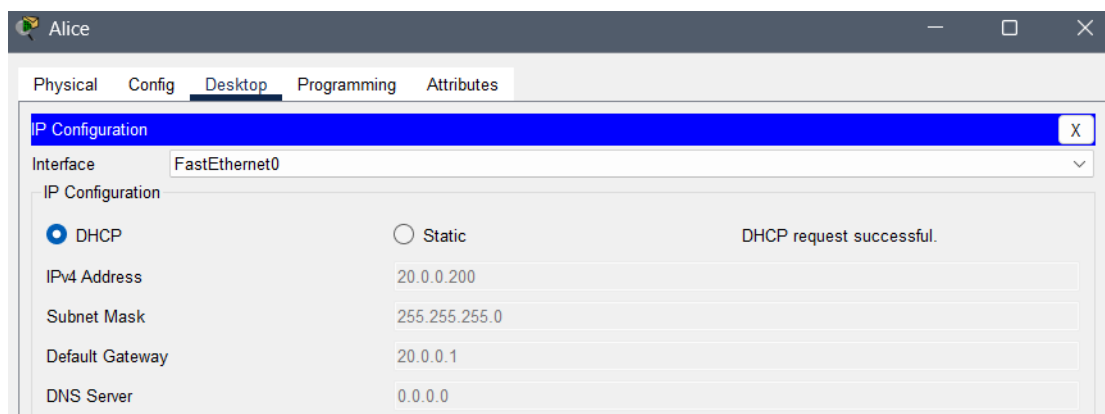
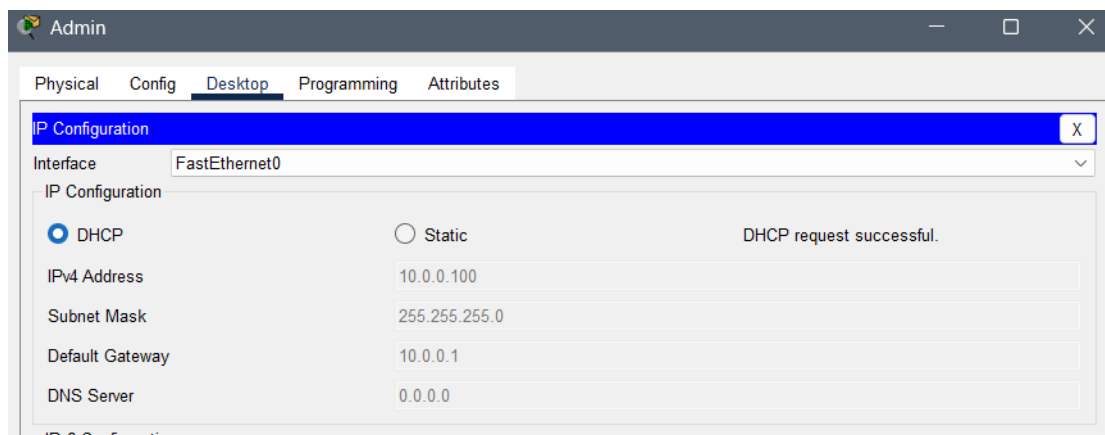
TFTP Server: 0.0.0.0

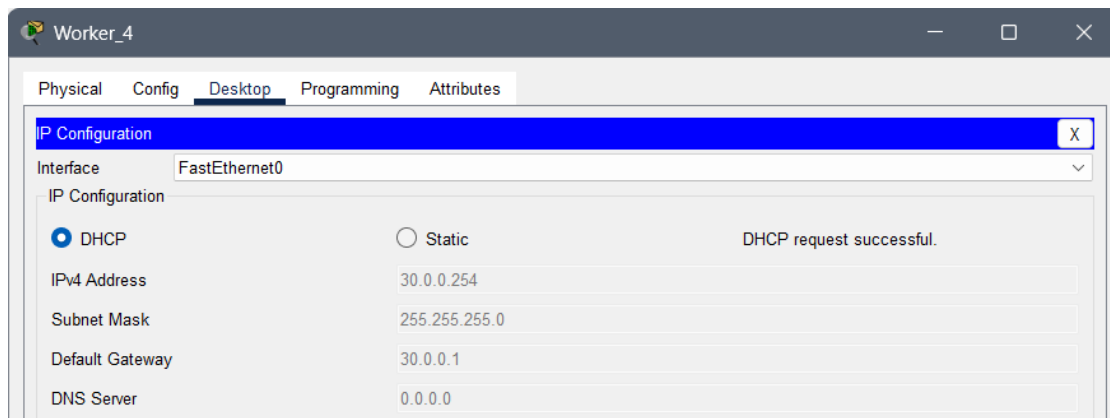
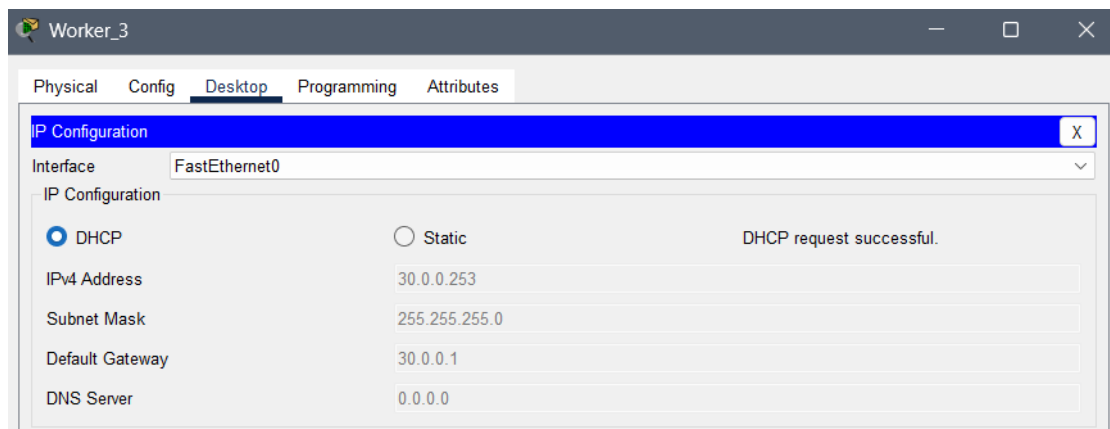
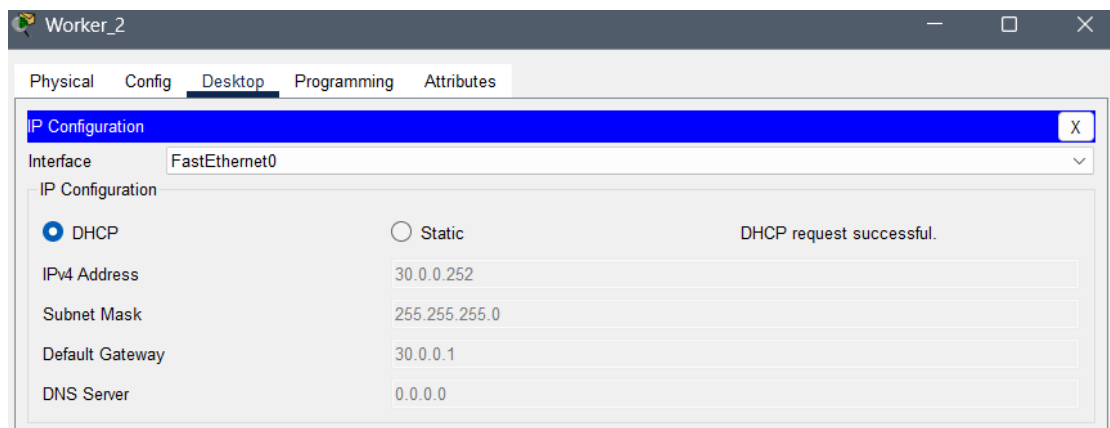
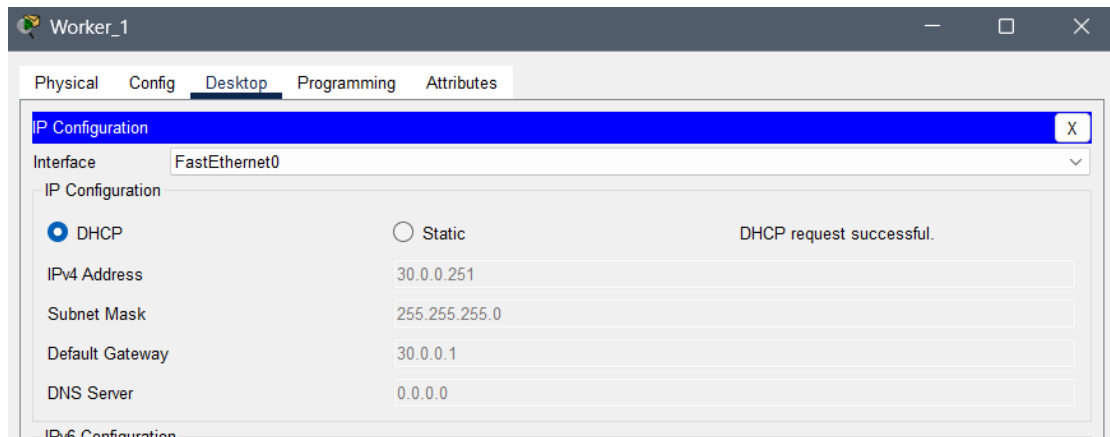
WLC Address: 0.0.0.0

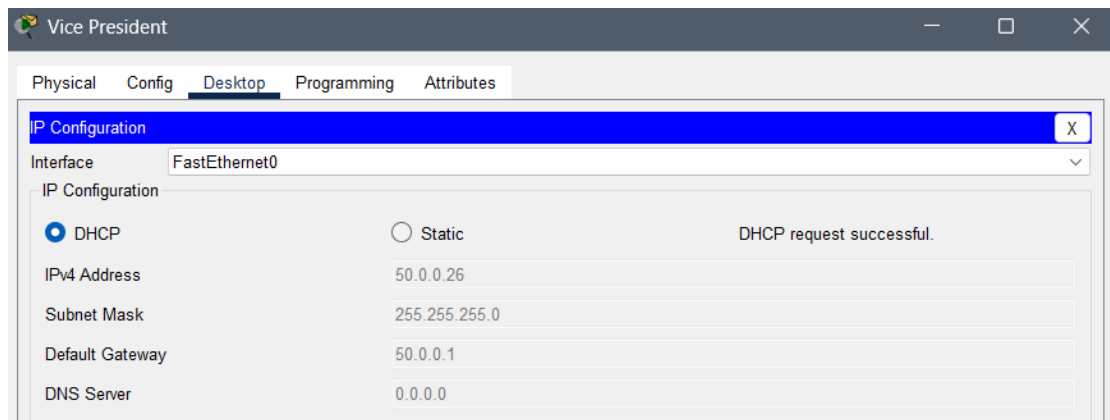
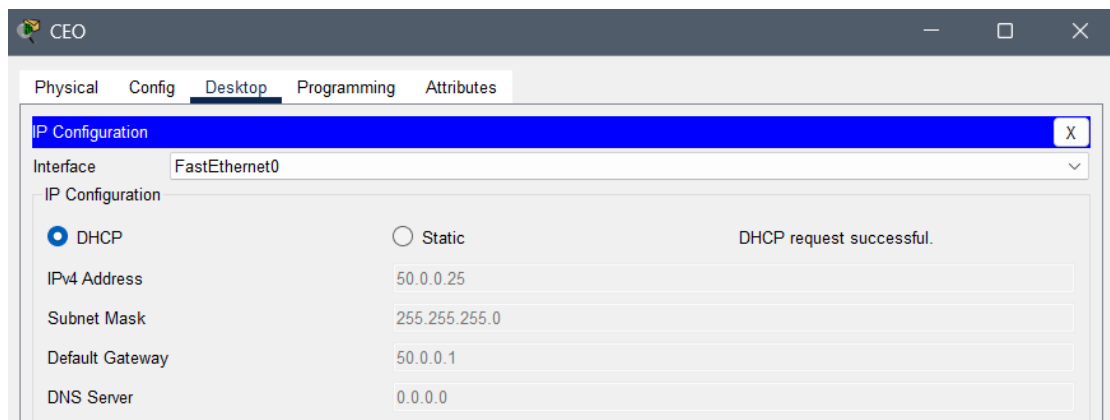
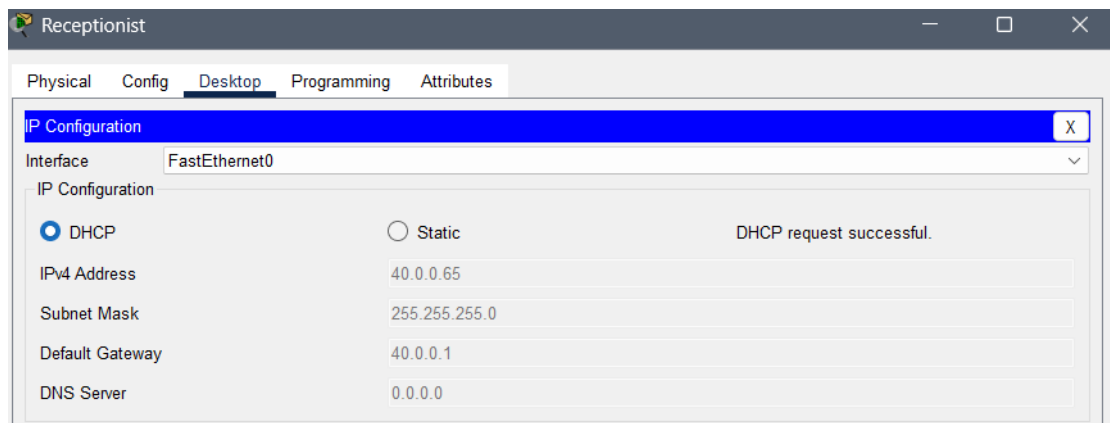
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
executivePool	50.0.0.1	0.0.0.0	50.0.0.25	255.255.255.0	10	0.0.0.0	0.0.0.0
receptionPool	40.0.0.1	0.0.0.0	40.0.0.65	255.255.255.0	10	0.0.0.0	0.0.0.0
workerPool	30.0.0.1	0.0.0.0	30.0.0.250	255.255.255.0	5	0.0.0.0	0.0.0.0
branchOfficePool	20.0.0.1	0.0.0.0	20.0.0.200	255.255.255.0	10	0.0.0.0	0.0.0.0
adminPool	10.0.0.1	0.0.0.0	10.0.0.100	255.255.255.0	10	0.0.0.0	0.0.0.0
serverPool	192.168.1.1	0.0.0.0	192.168.1.0	255.255.255.0	10	0.0.0.0	0.0.0.0

Step 6: set the PCs' ip address

To set the PCs' ip address we have to select each PC, click on Desktop> ip address and select the DHCP option. Then the IP will be configured by itself because of the step 5.

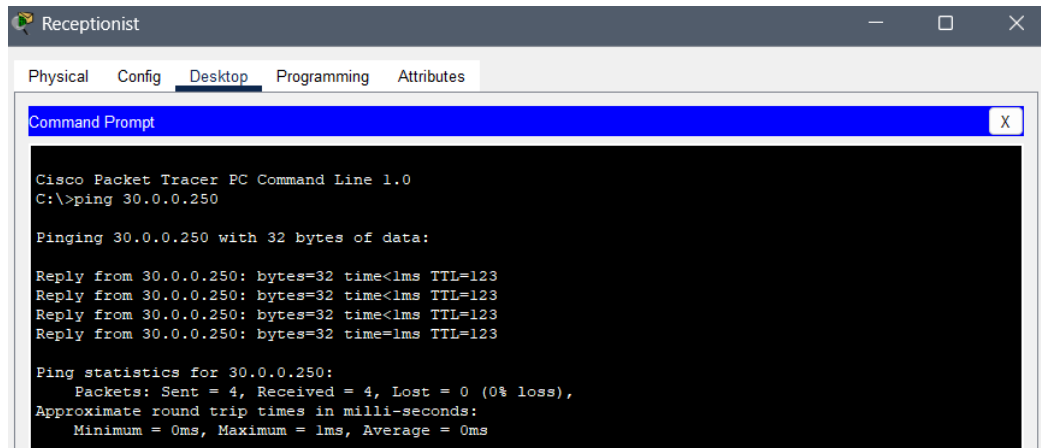






Test

Let's try to send a packet from the Receptionist to a worker. By using the command prompt from Receptionist's PC, we can achieve that.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.250

Pinging 30.0.0.250 with 32 bytes of data:

Reply from 30.0.0.250: bytes=32 time<1ms TTL=123
Reply from 30.0.0.250: bytes=32 time<1ms TTL=123
Reply from 30.0.0.250: bytes=32 time<1ms TTL=123
Reply from 30.0.0.250: bytes=32 time=1ms TTL=123

Ping statistics for 30.0.0.250:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

So, as it seems, everything works fine